

```
(gdb) backtrace
#0  syscall () at kernel/syscall.c:138
#1  0x0000000080001b2a in usertrap () at kernel/trap.c:68
#2  0x00000003ffffff09c in ?? ()
(gdb) □
```

```

Sep 23 19:12
saad@saad-dweller: ~/xv6-labs-2026 -- gdb-multiarch kernel/kernel
kernel/syscall.c
146 p->trapframe->e0 = syscalls[num]();
147 } else {
148     printf("id %x: unknown sys call %d\n", p->pid, p->name, num);
149     p->trapframe->e0 = -1;
150 }
151 }

remote thread 1.3 (src) in: syscall
#1  0x0000000000001b2a in usertrap () at kernel/trap.c:60
#2  0x0000003fffffff99c in ?? ()
(gdb) set logging off
Warning: 'set logging off', an alias for the command 'set logging enabled', is deprecated.
Use 'set logging enabled off'.
Done logging to /tmp/gdb-output.txt.
(gdb) set logging file /tmp/gdb-output.txt
(gdb) set logging on
Copying output to /tmp/gdb-output.txt.
Copying debug output to /tmp/gdb-output.txt.
(gdb) backtrace
#0  syscall () at kernel/syscall.c:138
#1  0x0000000000001b2a in usertrap () at kernel/trap.c:60
#2  0x0000003fffffff99c in ?? ()
(gdb) set logging off
Done logging to /tmp/gdb-output.txt.
(gdb)
L138 PC: 0x00001d0a
```

```
Sep 23 19:31
saad@sand-dweller: ~/xv6-labs-2026 -- gdb-multiarch kernel/kernel
~/xv6-labs-2026

kernel/syscall.c
122 SYS_sbrk = sys_sbrk,
123 SYS_pause = sys_pause,
124 SYS_uptime = sys_uptime,
125 SYS_open = sys_open,
126 SYS_write = sys_write,
127 SYS_mknod = sys_mknod,
128 SYS_unlink = sys_unlink,
129 SYS_link = sys_link,
130 SYS_mkdir = sys_mkdir,
131 SYS_close = sys_close,
132 SYS_sync = sys_sync,
133 // clang-format on
134 }
135
136 void
137 syscall(void)
138 {
139     int num;
140     struct proc *p = myproc();
141
142     num = p->trapframe->a7;
143     if (num > 0 && num < NELEM(syscalls) && syscalls[num]) {
144         // Use num to lookup the system call function for num, call it,
145         // and store its return value in p->trapframe->a0
146         p->trapframe->a0 = syscalls[num]();
147     } else {
148         printf("bad sys call %d\n", p->pid, p->name, num);
149         p->trapframe->a0 = -1;
150     }
151 }

remote thread 1.3 (src) in: syscall
(gdb) n
(gdb) p /x *p
$1 = {lock = {locked = 0x0, name = 0x00007130, cpu = 0x0}, state = 0x4, chan = 0x0, killed = 0x0, xstate = 0x0, pid = 0x1, parent = 0x0, kstack = 0x3fffffd000, sz = 0x4000, pagetable = 0x07f52000, trapframe = 0x07f56000, context = {ra = 0x0000135c, sp = 0x3fffffd000, s0 = 0x3fffffd000, s1 = 0x0000a740, s2 = 0x0000a310, s3 = 0x0, s4 = 0x3, s5 = 0x0001b3e0, s6 = 0x0, s7 = 0x100, s8 = 0xffffffffffffffff, s9 = 0x400, s10 = 0x2, s11 = 0x30}, ofile = {0x0 <repeats 10 times>}, cwd = 0x0010830, name = {0x09, 0x0e, 0x09, 0x74, 0x0 <repeats 12 times>}}
(gdb) p /x p->trapframe->a7
$2 = 0x11
(gdb) p /x *p
$3 = {lock = {locked = 0x0, name = 0x00007130, cpu = 0x0}, state = 0x4, chan = 0x0, killed = 0x0, xstate = 0x0, pid = 0x1, parent = 0x0, kstack = 0x3fffffd000, sz = 0x4000, pagetable = 0x07f52000, trapframe = 0x07f56000, context = {ra = 0x0000135c, sp = 0x3fffffd000, s0 = 0x3fffffd000, s1 = 0x0000a740, s2 = 0x0000a310, s3 = 0x0, s4 = 0x3, s5 = 0x0001b3e0, s6 = 0x0, s7 = 0x100, s8 = 0xffffffffffffffff, s9 = 0x400, s10 = 0x2, s11 = 0x30}, ofile = {0x0 <repeats 10 times>}, cwd = 0x0010830, name = {0x09, 0x0e, 0x09, 0x74, 0x0 <repeats 12 times>}}
(gdb) p p->trapframe->a7
$4 = 17
(gdb) p/x p->trapframe->a7
$5 = 0x11
(gdb) p /x $sstatus
$6 = 0x200000022
(gdb) []
```

Sep 23 19:40

saad@saad-dweller: ~/xv6-labs-2026 -- gdb-multiarch kernel/kernel

44%

```
0x00001d6a <syscall> addi    sp,sp,32
0x00001d6c <syscall+2> sd      ra,24(sp)
0x00001d6e <syscall+4> sd      a0,16(sp)
0x00001d70 <syscall+6> sd      s1,8(sp)
0x00001d72 <syscall+8> addi    a0,sp,32
0x00001d74 <syscall+10> jal      0x0000db0 <myproc>
0x00001d76 <syscall+12> mv       s1,a0
B> 0x00001d7a <syscall+16> lw       a3,0(zero) # 0x0
0x00001d7c <syscall+20> addiw    a4,a3,1
0x00001d7e <syscall+24> li        a3,21
0x00001d84 <syscall+26> bltu     a3,a4,0x00001da2 <syscall+50>
0x00001d86 <syscall+30> slll     a4,a3,0x3
0x00001d8e <syscall+34> auipc    a0,0x0
0x00001d90 <syscall+38> addi     a2,a0,-1132
0x00001d94 <syscall+42> add      a2,a2,a4
0x00001d96 <syscall+44> ld       a1,4(a0)
0x00001d98 <syscall+46> beqz     a5,0x00001da2 <syscall+56>
0x00001d9a <syscall+48> ld       s1,8(a0)
0x00001d9c <syscall+50> jalr     a0
0x00001d9e <syscall+52> sd      a0,112(s1)
0x00001da8 <syscall+54> j        0x00001dba <syscall+88>
0x00001da2 <syscall+56> addi     a2,a1,344
0x00001da6 <syscall+60> lw       a1,48(s1)
0x00001da8 <syscall+62> auipc    a0,0x5
0x00001dac <syscall+66> addi     a0,a0,1448
0x00001db0 <syscall+70> jal      0x0000530e <printf>
0x00001db4 <syscall+74> ld       a2,8(s1)
0x00001db6 <syscall+76> li        a1,1
0x00001db8 <syscall+78> sd      a1,112(a0)
0x00001dba <syscall+80> ld       ra,24(sp)
0x00001dbc <syscall+82> ld       s0,16(sp)
0x00001dbe <syscall+84> ld       s1,8(sp)
0x00001dc0 <syscall+86> addi     sp,sp,32
0x00001dc2 <syscall+88> ret
```

remote thread 1.1 (asm) in: syscall

(gdb)

L142 PC: 0x00001d7a

```

Sep 23 20:12
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
~/xv6-labs-2026

init: starting sh
$ sandbox 32768 - cat README
cat: cannot open README
$ cat README
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echtwermer, eyalz800, Nelson Elhage, Saar Ettinger, Alice Ferrazzi,
Nathaniel Filardo, fiespark, Peter Froehlich, Yakir Golan, Shivan
Handa, Matt Harvey, Bryan Henry, jaichenhengjie, Jim Huang, Matias
Kakay, John Jolly, Alexander Kapshuk, Anders Kaseorg, keha095,
Wolfgang Keller, Jungwoo Kim, Jonathan Kimball, Eddie Kohler, Vadim
Kolontsov, Austin Liew, Bruce Lowekamp, Iostman, Pavan Maddamsetti,
Inbar Marinescu, Yandong Mao, Matan Shabtay, Hitoshi Mitake, Carmi
Merinovich, mes900903, Mark Morrissey, mtasn, Joel Nider, Hayato Ohhashi,
OptimisticSide, pappapapa, phosphagos, Harry Porter, Greg Price, Zheng
ghuo, Quancheng, RayAndrew, Jude Rich, sogfault, Ayan Shafiqat, Eldar
Sehayek, Yongning Shen, Fumiya Shigenitsu, snoire, Tootie, Can Tenny,
tyfkda, Warren Toomey, Stephen Tu, Alissa Tung, Rafael Ubal, unicornx,
Anane Uehara, Pablo Ventura, Luc Videau, Xi Wang, WaheedHafez, Keiichi
Watanabe, Lucas Wolf, Nicolas Wolovick, wxdao, Grant Wu, x653, Andy
Zhang, Jindong Zhang, Icenowy Zheng, Zhuyi1997, and Zou Chang Wei.

ERROR REPORTS

Please send errors and suggestions to Frans Kaashoek and Robert Morris
(kaashoek,rmb@mit.edu). The main purpose of xv6 is as a teaching
operating system for MIT's 6.1810, so we are more interested in
simplifications and clarifications than new features.

BUILDING AND RUNNING XV6

You will need a RISC-V "newlib" tool chain from
https://github.com/riscv/riscv-gnu-toolchain, and qemu compiled for
riscv64-softmmu. Once they are installed, and in your shell
search path, you can run "make qemu".
$ sandbox 32768 - cat README
cat: cannot open README
$
```

```
Sep 23 20:14
saad@sand-dweller: ~/xv6-labs-2026
cat: cannot open README
$ sandbox 0 - cat README
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Jokay, John Jolly, Alexander Kapshuk, Anders Kaseorg, keha05,
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Marinovich, mesp00000, Mark Morrissey, ntsos, Joel Wider, Hayato Ohhashi,
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https://github.com/riscv/riscv-gnu-toolchain, and qemu compiled for
riscv64-softmmu. Once they are installed, and in your shell
search path, you can run "make qemu".
$ QEMU: Terminated
saad@sand-dweller:~/xv6-labs-2026$ ./grade-lab-syscall sandbox_mask
make: 'kernel/kernel' is up to date.
== Test sandbox_mask == sandbox_mask: OK (0.6s)
saad@sand-dweller:~/xv6-labs-2026$
```



```
Sep 23 22:19
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
~/xv6-labs-2026

no-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/dorphan.o user/dorphan.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_dorphan user/dorphan.o user/_lib.o user/_usys.o user/_printf.o user/_umalloc.o
riscv64-linux-gnu-objdump -S user/_dorphan > user/dorphan.asm
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_SYSCALL -DLAB_SYSCALL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/sync.o user/sync.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sync user/sync.o user/_lib.o user/_usys.o user/_printf.o user/_umalloc.o
riscv64-linux-gnu-objdump -S user/_sync > user/sync.asm
riscv64-linux-gnu-objdump -t user/_sync | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$d' > user/sync.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_SYSCALL -DLAB_SYSCALL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/sandbox.o user/sandbox.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sandbox user/sandbox.o user/_lib.o user/_usys.o user/_printf.o user/_umalloc.o
riscv64-linux-gnu-objdump -S user/_sandbox > user/sandbox.asm
riscv64-linux-gnu-objdump -t user/_sandbox | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$d' > user/sandbox.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_SYSCALL -DLAB_SYSCALL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/sbtest.o user/sbtest.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sbtest user/sbtest.o user/_lib.o user/_usys.o user/_printf.o user/_umalloc.o
riscv64-linux-gnu-objdump -S user/_sbtest > user/sbtest.asm
riscv64-linux-gnu-objdump -t user/_sbtest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$d' > user/sbtest.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_SYSCALL -DLAB_SYSCALL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/attack.o user/attack.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_attack user/attack.o user/_lib.o user/_usys.o user/_printf.o user/_umalloc.o
riscv64-linux-gnu-objdump -S user/_attack > user/attack.asm
riscv64-linux-gnu-objdump -t user/_attack | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$d' > user/attack.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_SYSCALL -DLAB_SYSCALL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-protector -fno-pie -no-pie -c -o user/secret.o user/secret.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_secret user/secret.o user/_lib.o user/_usys.o user/_printf.o user/_umalloc.o
riscv64-linux-gnu-objdump -S user/_secret > user/secret.asm
riscv64-linux-gnu-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$d' > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/cat user/echo user/forktest user/grep user/init user/kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/r/_forphan user/_dorphan user/_sync user/_sandbox user/_sbtest user/_attack user/_secret
mmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2868
balloc: first 1137 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6). xv6 loosely follows the structure and style of v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
(kaashoek,rtm@mit.edu)). The main purpose of xv6 is as a teaching
$ sandbox 32768 README grep xv6 x
grep: cannot open x
$
```

Sep 23 22:19

saad@sand-dweller: ~/xv6-labs-2026 — make qemu

~/xv6-labs-2026

```
Version 6 (v6).  xv6 loosely follows the structure and style of v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
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$ sandbox 32768 README grep xv6 x
grep: cannot open x
$ sandbox 64 README cat README
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ERROR REPORTS

Please send errors and suggestions to Frans Kaashoek and Robert Morris
(kaashoek,rtm@mit.edu).  The main purpose of xv6 is as a teaching
operating system for MIT's 6.0316, so we are more interested in
simplifications and clarifications than new features.

BUILDING AND RUNNING XV6

You will need a RISC-V "newlib" tool chain from
https://github.com/riscv/riscv-gnu-toolchain, and qemu compiled for
riscv64-softmmu.  Once they are installed, and in your shell
search path, you can run "make qemu".
$
```